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****************************************************
*                                                  *
*           STAAD.Pro                              *
*           Version  2006    Bld 1002.US           *
*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    NOV 16, 2007                  *
*           Time=    16:52:21                      *
*                                                  *
*      USER ID: roark                              *
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     1. STAAD SPACE BLDG -26(STEEL)
INPUT FILE: BLDG-26 STEEL Structure.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-SEPT-07
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. *************************************************************************
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 0 5.312 0.662; 3 0 6.317 0.503; 4 0 7.056 -0.14; 5 0 7.253 -1.111
    10. 6 0 7.317 -5.809; 7 0 7.347 -8.999; 8 0 7.306 -12.459; 9 0 7.253 -16.887
    11. 10 0 7.056 -17.856; 11 0 6.317 -18.5; 12 0 5.312 -18.659; 13 0 0 -17.997
    12. 14 6.5 0 0; 15 6.5 5.312 0.662; 16 6.5 6.317 0.503; 17 6.5 7.056 -0.14
    13. 18 6.5 7.253 -1.111; 19 6.5 7.317 -5.809; 20 6.5 7.347 -8.999
    14. 21 6.5 7.306 -12.459; 22 6.5 7.253 -16.887; 23 6.5 7.056 -17.856
    15. 24 6.5 6.317 -18.5; 25 6.5 5.312 -18.659; 26 6.5 0 -17.997; 27 12.5 0 0
    16. 28 12.5 5.312 0.662; 29 12.5 6.317 0.503; 30 12.5 7.056 -0.14
    17. 31 12.5 7.253 -1.111; 32 12.5 7.317 -5.809; 33 12.5 7.347 -8.999
    18. 34 12.5 7.306 -12.459; 35 12.5 7.253 -16.887; 36 12.5 7.056 -17.856
    19. 37 12.5 6.317 -18.5; 38 12.5 5.312 -18.659; 39 12.5 0 -17.997; 40 19 0 0
    20. 41 19 5.312 0.662; 42 19 6.317 0.503; 43 19 7.056 -0.14; 44 19 7.253 -1.111
    21. 45 19 7.317 -5.809; 46 19 7.347 -8.999; 47 19 7.306 -12.459
    22. 48 19 7.253 -16.887; 49 19 7.056 -17.856; 50 19 6.317 -18.5
    23. 51 19 5.312 -18.659; 52 19 0 -17.997; 53 20.5 7.056 -0.14
    24. 54 20.5 7.056 -17.856; 55 -1.5 7.056 -0.14; 56 -1.5 7.056 -17.856
    25. 57 22 7.317 -5.809; 58 22 7.306 -12.459; 59 -3 7.317 -5.809
    26. 60 -3 7.306 -12.459; 61 22.2 7.347 -8.999; 62 -3.2 7.347 -8.999
    27. 63 0 7.27428 -15.1088; 64 6.5 7.27428 -15.1088; 65 12.5 7.27428 -15.1088
    28. 66 19 7.27428 -15.1088; 67 0 7.28269 -3.2908; 68 6.5 7.28269 -3.2908
    29. 69 12.5 7.28269 -3.2908; 70 19 7.28269 -3.2908; 71 20.8 7.27428 -15.1088
    30. 72 -1.8 7.27428 -15.1088; 73 20.8 7.28269 -3.2908; 74 -1.8 7.28269 -3.2908
    31. 75 0 2.67927 0.333901; 76 6.5 2.67927 0.333901; 77 12.5 2.67927 0.333901
    32. 78 19 2.67927 0.333901; 79 20.3 2.67927 0.333901; 80 -1.3 2.67927 0.333901
    33. 81 0 2.67927 -18.3309; 82 6.5 2.67927 -18.3309; 83 12.5 2.67927 -18.3309
    34. 84 19 2.67927 -18.3309; 85 20.3 2.67927 -18.3309; 86 -1.3 2.67927 -18.3309
    35. 87 20.3 5.312 0.662; 88 20.3 5.312 -18.659; 89 -1.3 5.312 0.662
    36. 90 -1.3 5.312 -18.659; 91 9.7 7.28269 -3.2908; 93 9.5 7.347 -8.999
    37. 95 9.5 7.27428 -15.1088; 96 0 6.817 -5.809; 97 0 6.806 -12.459
    38. 98 6.5 6.817 -5.809; 99 6.5 6.806 -12.459; 100 12.5 6.817 -5.809
    39. 101 12.5 6.806 -12.459; 102 19 6.817 -5.809; 103 19 6.806 -12.459
    40. 104 0 6.856 -0.14; 105 0 6.856 -17.856; 106 6.5 6.856 -0.14
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41. 107 6.5 6.856 -17.856; 108 12.5 6.856 -0.14; 109 12.5 6.856 -17.856
42. 110 19 6.856 -0.14; 111 19 6.856 -17.856; 112 0 6.556 -0.14
43. 113 0 6.556 -17.856; 114 6.5 6.556 -0.14; 115 6.5 6.556 -17.856
44. 116 12.5 6.556 -0.14; 117 12.5 6.556 -17.856; 118 19 6.556 -0.14
45. 119 19 6.556 -17.856; 128 0 6.556 -18.2917; 129 6.5 6.556 -18.2917
46. 130 12.5 6.556 -18.2917; 131 19 6.556 -18.2917; 132 0 6.556 0.295047
47. 133 6.5 6.556 0.295047; 134 12.5 6.556 0.295047; 135 19 6.556 0.295047
48. *************************************************************************
49. MEMBER INCIDENCES
50. 1 1 75; 2 2 3; 3 3 132; 4 4 5; 5 5 67; 6 6 7; 7 7 8; 8 8 63; 9 10 9; 10 11 128
51. 11 12 11; 12 13 81; 13 14 76; 14 15 16; 15 16 133; 16 17 18; 17 18 68
52. 18 19 20; 19 20 21; 20 21 64; 21 23 22; 22 24 129; 23 25 24; 24 26 82
53. 25 27 77; 26 28 29; 27 29 134; 28 30 31; 29 31 69; 30 32 33; 31 33 34
54. 32 34 65; 33 36 35; 34 37 130; 35 38 37; 36 39 83; 37 40 78; 38 41 42
55. 39 42 135; 40 43 44; 41 44 70; 42 45 46; 43 46 47; 44 47 66; 45 49 48
56. 46 50 131; 47 51 50; 48 52 84; 52 63 9; 53 64 22; 54 65 35; 55 66 48; 56 67 6
57. 57 68 19; 58 69 32; 59 70 45; 95 75 2; 96 76 15; 97 77 28; 98 78 41; 99 81 12
58. 100 82 25; 101 83 38; 102 84 51; 201 53 43; 202 43 30; 203 30 17; 204 17 4
59. 205 4 55; 206 73 70; 207 70 69; 208 69 91; 209 68 67; 210 67 74; 211 57 45
60. 212 45 32; 213 32 19; 214 19 6; 215 6 59; 216 61 46; 217 46 33; 218 33 93
61. 220 7 62; 221 58 47; 222 47 34; 223 34 21; 224 21 8; 225 8 60; 226 71 66
62. 227 66 65; 228 65 95; 229 64 63; 230 63 72; 231 54 49; 232 49 36; 233 36 23
63. 234 23 10; 235 10 56; 236 91 68; 238 93 20; 240 95 64; 241 20 7; 301 79 78
64. 302 78 77; 303 77 76; 304 76 75; 305 75 80; 306 85 84; 307 84 83; 308 83 82
65. 309 82 81; 310 81 86; 311 87 41; 312 41 28; 313 28 15; 314 15 2; 315 2 89
66. 316 88 51; 317 51 38; 318 38 25; 319 25 12; 320 12 90; 501 55 67; 502 4 74
67. 503 59 7; 504 6 62; 505 62 8; 506 7 60; 507 72 10; 508 63 56; 509 43 73
68. 510 53 70; 511 45 61; 512 57 46; 513 46 58; 514 61 47; 515 71 49; 516 54 66
69. 522 17 91; 523 91 32; 524 30 91; 525 91 19; 530 19 93; 531 93 34; 532 32 93
70. 533 93 21; 538 34 95; 539 95 23; 540 36 95; 541 95 21; 542 97 8; 543 96 6
71. 544 99 21; 545 98 19; 546 101 34; 547 100 32; 548 103 47; 549 102 45
72. 550 105 10; 551 107 23; 552 109 36; 553 111 49; 554 104 4; 555 106 17
73. 556 108 30; 557 110 43; 558 113 128; 559 115 129; 560 117 130; 561 119 131
74. 562 128 10; 564 129 23; 566 130 36; 567 131 49; 568 118 135; 569 116 134
75. 570 114 133; 571 112 132; 572 132 4; 573 133 17; 574 134 30; 575 135 43
76. START GROUP DEFINITION
77. MEMBER
78. _COLUMN 1 TO 3 10 TO 15 22 TO 27 34 TO 39 46 TO 48 95 TO 102 562 564 566 567 -
79. 572 TO 575
80. _RAFTER 4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 59
81. _SEC_BEAM_ROOF 201 TO 218 220 TO 236 238 240 241
82. _SIDE_BEAM 301 TO 320
83. _BRACING 501 TO 516 522 TO 525 530 TO 533 538 TO 541
84. _STUBCOLUMN 542 TO 561 568 TO 571
85. END GROUP DEFINITION
86. *************************************************************************
87. DEFINE MATERIAL START
88. ISOTROPIC STEEL
89. E 2.05E+008
90. POISSON 0.3
91. DENSITY 78
92. ALPHA 1.2E-005
93. DAMP 0.03
94. END DEFINE MATERIAL
95. MEMBER PROPERTY BRITISH
96. ************************MAIN COLUMN**************************************
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 97. 1 12 13 24 25 36 37 48 TAPERED 0.3 0.006 0.35 0.2 0.01 0.2 0.01
 98. 95 TO 102 TAPERED 0.35 0.006 0.4 0.2 0.01 0.2 0.01
 99. 2 3 10 11 14 15 22 23 26 27 34 35 38 39 46 47 562 564 566 567 572 TO 574 -
100. 575 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
101. *************************MAIN RAFTER *************************************
102. 4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 58 -
103. 59 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
104. *************************SEC BEAM ****************************************
105. 201 TO 218 220 TO 236 238 240 241 TAPERED 0.3 0.006 0.3 0.15 0.01 0.15 0.01
106. *************************SIDE BEAM *************************************
107. 301 TO 320 TAPERED 0.25 0.006 0.25 0.25 0.012 0.25 0.012
108. *************************STUB COLUMN *************************************
109. 542 TO 549 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
110. 550 TO 561 568 TO 571 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
111. *550 TO 561 568 TO 571 PRIS YD 0.4 ZD 0.01
112. *************************BRACING ******************************************
113. 501 TO 516 522 TO 525 530 TO 533 538 TO 541 TABLE ST UA70X70X6
114. ***************************************************************************
115. CONSTANTS
116. BETA 90 MEMB 301 TO 320 542 TO 557
117. MATERIAL STEEL ALL
118. *************************************************************************
119. SUPPORTS
120. 1 13 14 26 27 39 40 52 PINNED
121. 96 TO 119 PINNED
122. *4 6 8 10 17 19 21 23 30 32 34 36 43 45 47 49 PINNED
123. *4 6 8 10 17 19 21 23 30 32 34 36 43 45 47 49 128 TO 134 PINNED
124. *************************************************************************
125. MEMBER RELEASE
126. *558 TO 561 568 TO 571 END MX MY MZ
127. *550 TO 557 END MX MY MZ
128. 203 204 208 209 213 214 218 223 224 228 229 233 234 241 303 304 308 309 313 -
129. 314 318 319 START MZ
130. 202 203 207 212 213 217 222 223 227 232 233 236 238 240 302 303 307 308 312 -
131. 313 317 318 END MZ
132. MEMBER TRUSS
133. 501 TO 516 522 TO 525 530 TO 533 538 TO 541
134. DEFINE UBC LOAD
135. ZONE 0.15 I 1 RWX 4.5 RWZ 4.5 STYP 4 CT 0.0853 PX 0.381 PZ 0.381 NA 1 NV 1
136. JOINT WEIGHT
137. 4 WEIGHT 39.57
138. 6 WEIGHT 50.32
139. 8 WEIGHT 45.126
140. 10 WEIGHT 41.672
141. 17 WEIGHT 54.606
142. 19 WEIGHT 49.012
143. 21 WEIGHT 43.508
144. 23 WEIGHT 57.012
145. 30 WEIGHT 54.606
146. 32 WEIGHT 49.012
147. 34 WEIGHT 43.508
148. 36 WEIGHT 57.29
149. 43 WEIGHT 39.57
150. 45 WEIGHT 50.32
151. 47 WEIGHT 45.126
152. 49 WEIGHT 41.537
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153. LOAD 21 EQX (EARTHQUAKE LOAD )
154. UBC LOAD X
155. LOAD 22 EQZ (EARTHQUAKE LOAD )
156. UBC LOAD Z
157. LOAD 1 SELFWEIGHT
158. SELFWEIGHT Y -1
159. LOAD 2 DL
160. **DEAD LOAD = 0.8 KN/M^2
161. **BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
162. **DL1 = 3.3X0.8 =2.66=2.7 KN/M
163. **DL2 = 3.0X0.8 =2.4 KN/M
164. **DL3 = 2.7X0.8 =2.16=2.2 KN/M
165. MEMBER LOAD
166. 216 TO 218 220 238 241 301 TO 310 UNI GY -2.7
167. 211 TO 215 221 TO 225 UNI GY -2.4
168. 206 TO 210 226 TO 230 236 240 UNI GY -2.2
169. 201 TO 205 231 TO 235 311 TO 320 UNI GY -2.2
170. ***************************************
171. LOAD 3 LL
172. MEMBER LOAD
173. **LIVE LOAD = 0.6 KN/M^2
174. **BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
175. **LL1 = 3.3X0.6 =1.98=2.0 KN/M
176. **LL2 = 3.0X0.6 =1.8 KN/M
177. **LL3 = 2.7X0.6 =1.62=1.65 KN/M
178. 216 TO 218 220 238 241 UNI GY -2
179. 211 TO 215 221 TO 225 UNI GY -1.8
180. 201 TO 210 226 TO 236 240 UNI GY -1.65
181. **********A/C EQUIPMENT S/W = 5 KN ( POINT LOAD )
182. *JOINT LOAD
183. *20 33 FY -5
184. LOAD 4 PH(X-DIR)
185. ************************
186. **PH =1.5% (1.4DL+1.6LL)
187. JOINT LOAD
188. 4 FX 0.88
189. 6 FX 1.47
190. 8 FX 1.4
191. 10 FX 0.9
192. 17 FX 1.33
193. 19 FX 1.43
194. 21 FX 1.35
195. 23 FX 1.37
196. 30 FX 1.33
197. 32 FX 1.43
198. 34 FX 1.35
199. 36 FX 1.37
200. 43 FX 0.88
201. 45 FX 1.47
202. 47 FX 1.4
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49 FX 0.91
LOAD 5 PH(Z-DIR)
***********************
**PH =1.5% (1.4DL+1.6LL)
JOINT LOAD
4 FZ 0.88
6 FZ 1.47
8 FZ 1.4
10 FZ 0.9
17 FZ 1.33
19 FZ 1.43
21 FZ 1.35
23 FZ 1.37
30 FZ 1.33
32 FZ 1.43
34 FZ 1.35
36 FZ 1.37
43 FZ 0.88
45 FZ 1.47
47 FZ 1.4
49 FZ 0.91
********************************************************
LOAD 11 WL1 (0 DEG,INTERNAL SUCTION (CPI=-0.3)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=0.77,CPE(LEEWARD WALL=-0.5)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
MEMBER LOAD
306 TO 310 UNI GZ 5.31
221 TO 225 UNI Y 1.79
211 TO 215 UNI Y -1.86
301 TO 305 UNI GZ 0.99
316 TO 320 UNI GZ 3.32
311 TO 315 UNI GZ 0.62
226 TO 235 240 UNI Y 1.58
201 TO 210 236 UNI Y -1.65
216 TO 218 220 238 241 UNI Y -0.04
**********************************************************
LOAD 12 WL2 (0 DEG ,INTERNAL PRESSURE CPI=+0.2)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF)=-0.2
**CPE(WINDWARD WALL)=+0.77,CPE(LEEWARD WALL)=-0.5
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M
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**WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5-0.2)X4.0 =3.48 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5-0.2)X2.5 =2.17 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =3.65(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.2-0.2)X3.0 =1.65(UP) KN/M
**WL8(LEEWARD  ROOF)= 1.24(-0.2-0.2)X2.66 =1.49(UP)KN/M
MEMBER LOAD
306 TO 310 UNI GZ 2.83
301 TO 305 UNI GZ 3.48
316 TO 320 UNI GZ 1.77
311 TO 315 UNI GZ 2.17
221 TO 225 UNI Y 3.65
211 TO 215 UNI Y 1.49
226 TO 235 240 UNI Y 3.24
201 TO 210 236 UNI Y 1.32
216 TO 218 220 238 241 UNI Y 2.83
********************************************************
LOAD 13 WL3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=0.2,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=-0.8,CPE(LEEWARD WALL=-0.8)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(-0.8+0.3)X4.0 =2.48 KN/M
**WL2(WINDWARD WALL)= 1.24(-0.8+0.3)X2.5 =1.55 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.8+0.3)X4.0 =2.48 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.8+0.3)X2.5 =1.55 KN/M
**WL5(WINDWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL6(WINDWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
MEMBER LOAD
306 TO 310 UNI GZ -2.48
301 TO 305 UNI GZ 2.48
316 TO 320 UNI GZ -1.55
311 TO 315 UNI GZ 1.55
221 TO 225 UNI Y -1.86
211 TO 215 UNI Y -1.86
226 TO 235 240 UNI Y -1.65
201 TO 210 236 UNI Y -1.65
216 TO 218 220 238 241 UNI Y -2.05
********************************************************
LOAD 14 WL4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.7,CPE(LEEWARD ROOF=-0.7)
**CPE(WINDWARD WALL)=-0.8,CPE(LEEWARD WALL=-0.8)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(-0.8-0.2)X4.0 =4.97 KN/M
**WL2(WINDWARD WALL)= 1.24(-0.8-0.2)X2.5 =3.1 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.8-0.2)X4.0 =4.97 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.8-0.2)X2.5 =3.1 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.7-0.2)X3.0 =3.35(UP) KN/M
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**WL6(WINDWARD ROOF)= 1.24(-0.7-0.2)X2.66 =2.97(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.7-0.2)X3.0 =3.35(UP) KN/M
**WL8(LEEWARD  ROOF)= 1.24(-0.7-0.2)X2.66=2.97(UP)KN/M
MEMBER LOAD
306 TO 310 UNI GZ -4.97
301 TO 305 UNI GZ 4.97
316 TO 320 UNI GZ -3.1
311 TO 315 UNI GZ 3.1
221 TO 225 UNI Y 3.35
211 TO 215 UNI Y 3.35
226 TO 235 240 UNI Y 2.97
201 TO 210 236 UNI Y 2.97
216 TO 218 220 238 241 UNI Y 3.69
******************************************************************
LOAD 15 WL5 (0 DEG ,INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)
MEMBER LOAD
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF=0.2)
**CPE(WINDWARD WALL)=0.77,CPE(LEEWARD WALL=-0.5)
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
**WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
**WL8(LEEWARD  ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
306 TO 310 UNI GZ -0.99
301 TO 305 UNI GZ -5.31
316 TO 320 UNI GZ -0.62
311 TO 315 UNI GZ -3.32
221 TO 225 UNI Y -1.86
211 TO 215 UNI Y 1.79
226 TO 235 240 UNI Y -1.65
201 TO 210 236 UNI Y 1.58
216 218 220 238 UNI Y -0.04
241 UNI Y -0.04
217 UNI Y -0.04
217 UNI Y -0.04
*******************************************************************
LOAD 16 WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)
MEMBER LOAD
**WIND LOAD(Q) = 1.24 KN/M^2
**CPE(WINDWARD ROOF)=-0.78,CPE(LEEWARD ROOF)=-0.2
**CPE(WINDWARD WALL)=+0.77,CPE(LEEWARD WALL)=-0.5
**BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
**BEAM SPACING (WALL) = 4.0M , 2.5 M
**WIND LOAD = QX(CPE-CPI)XBEAM SPACING
**WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M
**WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
**WL3(LEEWARD  WALL)= 1.24(-0.5-0.2)X4.0 =3.48 KN/M
**WL4(LEEWARD  WALL)= 1.24(-0.5-0.2)X2.5 =2.17 KN/M
**WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =3.65(UP) KN/M
```



```
**WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
**WL7(LEEWARD  ROOF)= 1.24(-0.2-0.2)X3.0 =1.65(UP) KN/M
**WL8(LEEWARD  ROOF)= 1.24(-0.2-0.2)X2.66 =1.49(UP)KN/M
306 TO 310 UNI GZ -3.48
301 TO 305 UNI GZ -2.83
316 TO 320 UNI GZ -2.17
311 TO 315 UNI GZ -1.77
221 TO 225 UNI Y 1.49
211 TO 215 UNI Y 3.65
226 TO 235 240 UNI Y 1.32
201 TO 210 236 UNI Y 3.24
216 218 220 238 UNI Y 2.83
241 UNI Y 2.83
217 UNI Y 2.83
217 UNI Y 2.83
*********************************************
**          SERVICEBILITY*                 **
*********************************************
LOAD COMB 101 1.0DL+1.0LL
1 1.0 2 1.0 3 1.0
LOAD COMB 102 1.0DL+0.5LL
1 1.0 2 1.0 3 0.5
LOAD COMB 103 0.8DL+0.8LL+0.8WL1
1 0.8 2 0.8 3 0.8 11 0.8
LOAD COMB 104 1.0DL+1.0WL1
1 1.0 2 1.0 11 1.0
LOAD COMB 105 0.8DL+0.8LL+0.8WL2
1 0.8 2 0.8 3 0.8 12 0.8
LOAD COMB 106 1.0DL+1.0WL2
1 1.0 2 1.0 12 1.0
LOAD COMB 107 0.8DL+0.8LL+0.8WL3
1 0.8 2 0.8 3 0.8 13 0.8
LOAD COMB 108 1.0DL+1.0WL3
1 1.0 2 1.0 13 1.0
LOAD COMB 109 0.8DL+0.8LL+0.8WL4
1 0.8 2 0.8 3 0.8 14 0.8
LOAD COMB 110 1.0DL+1.0WL4
1 1.0 2 1.0 14 1.0
LOAD COMB 111 0.8DL+0.8LL+0.8WL5
1 0.8 2 0.8 3 0.8 15 0.8
LOAD COMB 112 1.0DL+1.0WL4
1 1.0 2 1.0 15 1.0
LOAD COMB 113 0.8DL+0.8LL+0.8WL6
1 0.8 2 0.8 3 0.8 16 0.8
LOAD COMB 114 1.0DL+1.0WL6
1 1.0 2 1.0 16 1.0
*********EQX  &  EQZ  ************************************
**DL+EQL/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQL=EQX+0.3EQZ
**0.9 DL+EQL/1.4 =0.9 DL+0.72 EQX+0.22EQZ
**DL 0.75(LL+(EQL/1.4)) =1.0 DL+0.75 LL+0.54 EQX+0.17EQZ
***********************************************************
LOAD COMB 115 DL+0.72EQX+0.22EQZ
1 1.0 2 1.0 21 0.72 22 0.22
LOAD COMB 116 DL+0.72EQX-0.22EQZ
1 1.0 2 1.0 21 0.72 22 -0.22
LOAD COMB 117 DL-0.72EQX+0.22EQZ
```

```
1 1.0 2 1.0 21 -0.72 22 0.22
LOAD COMB 118 DL-0.72EQX+0.22EQZ
1 1.0 2 1.0 21 -0.72 22 -0.22
LOAD COMB 119 DL+0.72EQZ+0.22EQX
1 1.0 2 1.0 22 0.72 21 0.22
LOAD COMB 120 DL+0.72EQZ-0.22EQX
1 1.0 2 1.0 22 0.72 21 -0.22
LOAD COMB 121 DL-0.72EQZ+0.22EQX
1 1.0 2 1.0 22 -0.72 21 0.22
LOAD COMB 122 DL-0.72EQZ-0.22EQX
1 1.0 2 1.0 22 -0.72 21 -0.22
LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
1 0.9 2 0.9 21 0.72 22 0.22
LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
1 0.9 2 0.9 21 0.72 22 -0.22
LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
1 0.9 2 0.9 21 -0.72 22 0.22
LOAD COMB 126 0.9DL-0.72EQX+0.22EQZ
1 0.9 2 0.9 21 -0.72 22 -0.22
LOAD COMB 127 0.9DL+0.72EQZ+0.22EQX
1 0.9 2 0.9 22 0.72 21 0.22
LOAD COMB 128 0.9DL+0.72EQZ-0.22EQX
1 0.9 2 0.9 22 0.72 21 -0.22
LOAD COMB 129 0.9DL-0.72EQZ+0.22EQX
1 0.9 2 0.9 22 -0.72 21 0.22
LOAD COMB 130 0.9DL-0.72EQZ-0.22EQX
1 0.9 2 0.9 22 -0.72 21 -0.22
LOAD COMB 131 1.0DL+0.75LL+0.54EQX+0.17 EQZ
1 1.0 2 1.0 3 0.75 21 0.54 22 0.17
LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
1 1.0 2 1.0 3 0.75 21 0.54 22 -0.17
LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
1 1.0 2 1.0 3 0.75 21 -0.54 22 0.17
LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
1 1.0 2 1.0 3 0.75 21 -0.54 22 -0.17
LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
1 1.0 2 1.0 3 0.75 22 0.54 21 0.17
LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
1 1.0 2 1.0 3 0.75 22 0.54 21 -0.17
LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
1 1.0 2 1.0 3 0.75 22 -0.54 21 0.17
LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
1 1.0 2 1.0 3 0.75 22 -0.54 21 -0.17
***************************************************
**              STEEL DESIGN                     **
**************************************************
LOAD COMB 200 1.4DL+1.6LL
1 1.4 2 1.4 3 1.6
*******WL1***************************************
LOAD COMB 201 1.2DL+1.2LL+1.2WL1
1 1.2 2 1.2 3 1.2 11 1.2
LOAD COMB 202 1.4DL+1.4WL1
1 1.4 2 1.4 11 1.4
LOAD COMB 203 1DL+1.4WL1
1 1.0 2 1.0 11 1.4
**********WL2***********************
```



```
LOAD COMB 204 1.2DL+1.2LL+1.2WL2
1 1.2 2 1.2 3 1.2 12 1.2
LOAD COMB 205 1.4DL+1.4WL2
1 1.4 2 1.4 12 1.4
LOAD COMB 206 1DL+1.4WL2
1 1.0 2 1.0 12 1.4
*******WL3************************
LOAD COMB 207 1.2DL+1.2LL+1.2WL3
1 1.2 2 1.2 3 1.2 13 1.2
LOAD COMB 208 1.4DL+1.4WL3
1 1.4 2 1.4 13 1.4
LOAD COMB 209 1DL+1.4WL3
1 1.0 2 1.0 13 1.4
*******WL4************************
LOAD COMB 210 1.2DL+1.2LL+1.2WL4
1 1.2 2 1.2 3 1.2 14 1.2
LOAD COMB 211 1.4DL+1.4WL4
1 1.4 2 1.4 14 1.4
LOAD COMB 212 1DL+1.4WL4
1 1.0 2 1.0 14 1.4
*******WL5************************
LOAD COMB 213 1.2DL+1.2LL+1.2WL5
1 1.2 2 1.2 3 1.2 15 1.2
LOAD COMB 214 1.4DL+1.4WL5
1 1.4 2 1.4 15 1.4
LOAD COMB 215 1DL+1.4WL5
1 1.0 2 1.0 15 1.4
*******WL6************************
LOAD COMB 216 1.2DL+1.2LL+1.2WL6
1 1.2 2 1.2 3 1.2 16 1.2
LOAD COMB 217 1.4DL+1.4WL6
1 1.4 2 1.4 16 1.4
LOAD COMB 218 1DL+1.4WL6
1 1.0 2 1.0 16 1.4
*********EQX  &  EQZ  *************************
LOAD COMB 219 1.2DL+0.5LL+1.0EQX+0.3 EQZ
1 1.2 2 1.2 3 0.5 21 1.0 22 0.3
LOAD COMB 220 1.2DL+0.5LL+1.0EQX-0.3 EQZ
1 1.2 2 1.2 3 0.5 21 1.0 22 -0.3
LOAD COMB 221 1.2DL+0.5LL-1.0EQX+0.3 EQZ
1 1.2 2 1.2 3 0.5 21 -1.0 22 0.3
LOAD COMB 222 1.2DL+0.5LL-1.0EQX-0.3 EQZ
1 1.2 2 1.2 3 0.5 21 -1.0 22 -0.3
LOAD COMB 223 1.2DL+0.5LL+1.0EQZ+0.3 EQX
1 1.2 2 1.2 3 0.5 22 1.0 21 0.3
LOAD COMB 224 1.2DL+0.5LL+1.0EQZ-0.3 EQX
1 1.2 2 1.2 3 0.5 22 1.0 21 -0.3
LOAD COMB 225 1.2DL+0.5LL-1.0EQZ+0.3 EQX
1 1.2 2 1.2 3 0.5 22 -1.0 21 0.3
LOAD COMB 226 1.2DL+0.5LL-1.0EQZ-0.3 EQX
1 1.0 2 1.2 3 0.5 22 -1.0 21 -0.3
LOAD COMB 227 0.9DL+1.0EQX+0.3EQZ
1 0.9 2 0.9 21 1.0 22 0.3
LOAD COMB 228 0.9DL+1.0EQX-0.3EQZ2
1 0.9 2 0.9 21 1.0 22 -0.3
LOAD COMB 229 0.9DL-1.0EQX+0.3EQZ
```

```
539. 1 0.9 2 0.9 21 -1.0 22 0.3
540. LOAD COMB 230 0.9DL-1.0EQX+0.3EQZ
541. 1 0.9 2 0.9 21 -1.0 22 -0.3
542. LOAD COMB 231 0.9DL+1.0EQZ+0.3EQX
543. 1 0.9 2 0.9 22 1.0 21 0.3
544. LOAD COMB 232 0.9DL+1.0EQZ-0.3EQX
545. 1 0.9 2 0.9 22 1.0 21 -0.3
546. LOAD COMB 233 0.9DL-1.0EQZ+0.3EQX
547. 1 0.9 2 0.9 22 -1.0 21 0.3
548. LOAD COMB 234 0.9DL-1.0EQZ-0.3EQX
549. 1 0.9 2 0.9 22 -1.0 21 -0.3
550. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
551. 1 1.0 2 1.0 3 1.0 4 1.0
552. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
553. 1 1.0 2 1.0 3 1.0 5 1.0
554. ******DEFLECTION******************************
555. LOAD COMB 301 SELF+DL
556. 1 1.0 2 1.0
557. LOAD COMB 302 LL
558. 3 1.0
559. LOAD COMB 303 0.8(LL+WL1)
560. 3 0.8 11 0.8
561. LOAD COMB 304 0.8(LL+WL2)
562. 3 0.8 12 0.8
563. LOAD COMB 305 0.8(LL+WL3)
564. 3 0.8 13 0.8
565. LOAD COMB 306 0.8(LL+WL4)
566. 3 0.8 14 0.8
567. LOAD COMB 307 0.8(LL+WL5)
568. 3 0.8 15 0.8
569. LOAD COMB 308 0.8(LL+WL6)
570. 3 0.8 16 0.8
571. LOAD COMB 309 0.8(LL+EQX/1.4+0.3EQZ/1.4)
572. 3 0.8 21 0.57 22 0.17
573. LOAD COMB 310 0.8(LL+EQX/1.4-0.3EQZ/1.4)
574. 3 0.8 21 0.57 22 -0.17
575. LOAD COMB 311 0.8(LL-EQX/1.4+0.3EQZ/1.4)
576. 3 0.8 21 -0.57 22 0.17
577. LOAD COMB 312 0.8(LL-EQX/1.4-0.3EQZ/1.4)
578. 3 0.8 21 -0.57 22 -0.17
579. LOAD COMB 313 0.8(LL+EQZ/1.4+0.3EQX/1.4)
580. 3 0.8 22 0.57 21 0.17
581. LOAD COMB 314 0.8(LL+EQZ/1.4-0.3EQX/1.4)
582. 3 0.8 22 0.57 21 -0.17
583. LOAD COMB 315 0.8(LL-EQZ/1.4+0.3EQX/1.4)
584. 3 0.8 22 -0.57 21 0.17
585. LOAD COMB 316 0.8(LL-EQZ/1.4-0.3EQX/1.4)
586. 3 0.8 22 -0.57 21 -0.17
587. ************************************************
588. PERFORM ANALYSIS
```



```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   125/   182/    32
   ORIGINAL/FINAL BAND-WIDTH=   119/    15/     81 DOF
   TOTAL PRIMARY LOAD CASES =   13, TOTAL DEGREES OF FREEDOM =    654
   SIZE OF STIFFNESS MATRIX =       53 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     13.0/  23735.8 MB


**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.


 ************************************************************
 *                                                          *
 *  X  DIRECTION : Ta =  0.381 Tb =  0.115 Tuser =  0.381  *
 *  T =  0.381, LOAD FACTOR =  1.000                        *
 * UBC TYPE = 97                                            *
 * UBC FACTOR V = 0.1222 x        761.80 =      93.11 KN   *
 *                                                          *
 ************************************************************


 ************************************************************
 *                                                          *
 *  Z  DIRECTION : Ta =  0.381 Tb =  0.028 Tuser =  0.381  *
 *  T =  0.381, LOAD FACTOR =  1.000                        *
 * UBC TYPE = 97                                            *
 * UBC FACTOR V = 0.1222 x        761.80 =      93.11 KN   *
 *                                                          *
 ************************************************************


 589. LOAD LIST 101 TO 138
 590. PRINT SUPPORT REACTION
SUPPORT  REACTION
```